INCH - POUND

MIL-F-60445C
3 June 1991
SUPERSEDING
MIL-F-60445B(MU)
6 APRIL 1972

MILITARY SPECIFICATION
FIN AND NOZZLE ASSEMBLY FOR
ROCKET MOTOR 2.75 INCH MARK 4 AND MARK 40
PARTS FOR

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings, for two types of nozzle and fin assemblies designated as Fin and Nozzle Assembly For Rocket Motor 2.75 Inch Mark 4 and Mark 40.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340
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SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-way or 4-way (Partial)
QQ-S-781	-	Strapping, Steel and Seals
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tapes, Gummed Paper, Reinforced and Plain for Sealing and Securing

MILITARY

MIL-A-2550	-	Ammunition, General Specification for

STANDARDS

FEDERAL

FED-STD-151	-	Federal Test Method Standard Metals: Test Methods

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Ammunition Lot Numbering
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunitions Components
MIL-STD-1949	-	Inspection, Magnetic Particle
MIL-STD-6866	-	Inspection, Liquid Penetrant
MS171590	-	Pin Spring, Corrosion Resistant Steel

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

U. S. Naval Ordnance Test Station

1517508	-	Nozzle and Fin Assembly

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PUBLICATIONS

ET-1517508-1,-2 - Equipment Tabulation

(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 7207 - Spring Pins, Tubular Corrosion and Moderate Heat Resistant Steel

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. Unless otherwise specified, dimensions and tolerances of materials and component parts shall be in accordance with specified drawings and conditions specified herein.

3.3 Nozzle and fin assembly (see drawing 1517508). The nozzle and fin assembly shall comply with all requirements specified on drawing 1517508 and with all requirements specified in this specification.

3.4 Metal parts and other material. The metal parts (except the nozzle plate which shall meet the requirements of 3.5) and other material used shall be free from defects, including all surface imperfections and discontinuities and all internal defects in the finished component resulting from flaws or lack of soundness in the materials used.

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3.4.1 Surface imperfections. Surface imperfections include pits, seams, laps, tears, cracks, scale, blisters, cold shuts, and other defects of similar type.

3.4.2 Internal defects. Internal defects include pipes, lamination, segregations, internal cracks, porosity, flakes, inclusions, and other defects of similar type.

3.5 Magnetic particle inspection (see drawing 656925). When the nozzle plate is subjected to magnetic particle inspection, the following requirements shall be met.

3.5.1 Inclusions of sulfides or other ferrous compounds. Inclusions of sulfide or other ferrous compounds precipitated and rolled into the material:

a. To any degree in the thin section between the counterbored holes or adjacent to the central hole are not acceptable and the part shall be rejected. Shallow discontinuities in other areas of this surface of these holes are acceptable where they are backed up by a no less than 0.1875 inch thickness of sound material (see figure 1).

b. On either the inside or outside surface of the piston cylinder are unacceptable (see figure 1).

c. In the skirt that show on both the inside and outside surfaces are not acceptable, and the part shall be rejected. Shallow discontinuities in either surface that do not show on the other surface are acceptable (see figures 1 and 2).

3.5.2 Cracks or cracklike indications. Cracks and cracklike indications are not allowed except in the following areas:

a. On the surfaces adjacent to the detent groove not greater than 0.062 inch in depth.

b. On the outside surface of the skirt that does not extend down into the grooves more than 0.062 inch. No cracks are allowed on the "R" surface.

3.6 Pivot pin push test. The pivot pin shall withstand a static push of 20 pounds applied successively to each end without relative movement between the spring pin and the nozzle plate.

3.7 Insulation integrity. The nozzle and fin assembly, prior to installation of the shorting clip must measure not less than 10 megohms when measured with 100 volts between the contact disc and the bourrelet on the nozzle plate.

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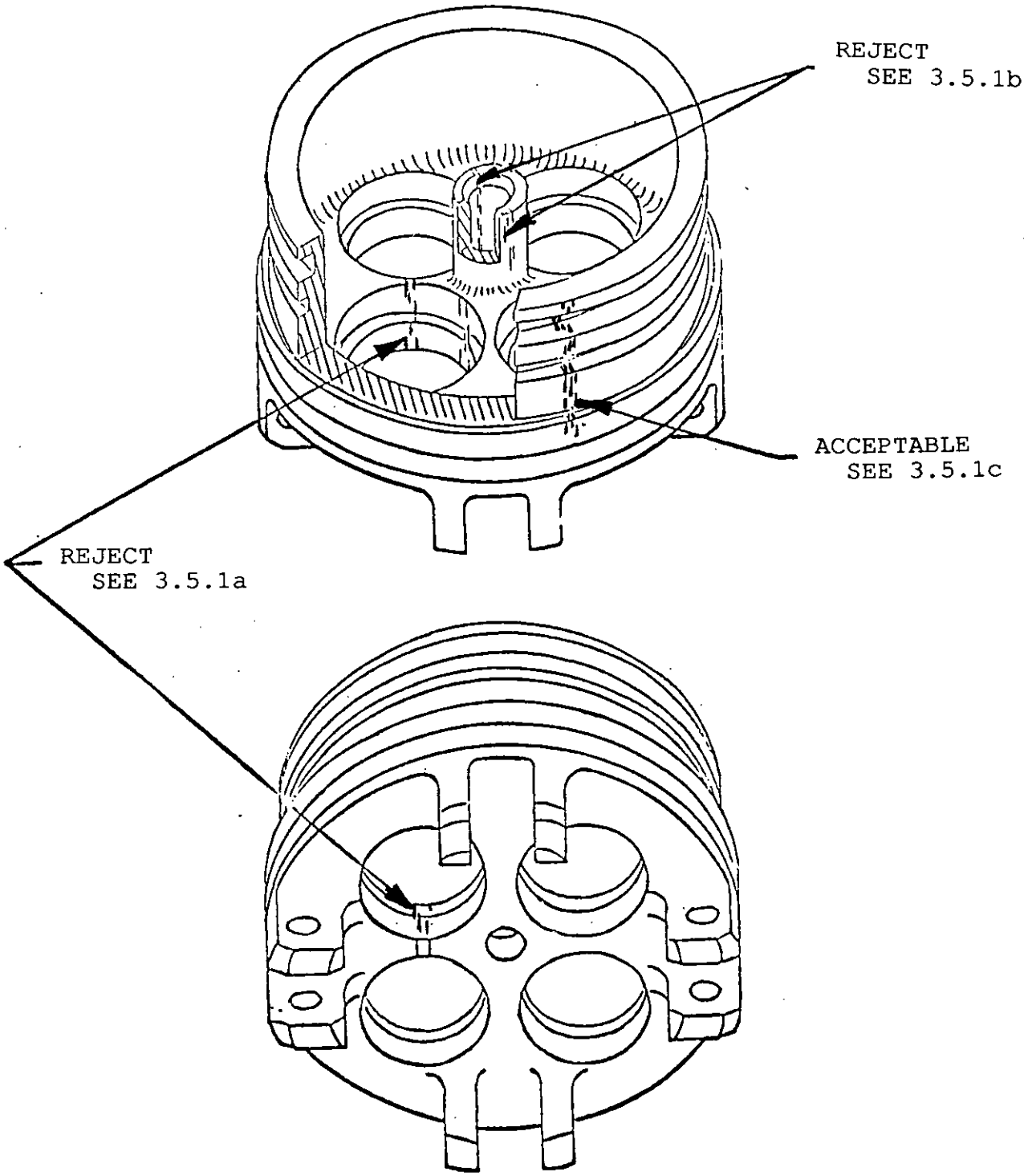


FIGURE 1. Discontinuities in piston cylinder.

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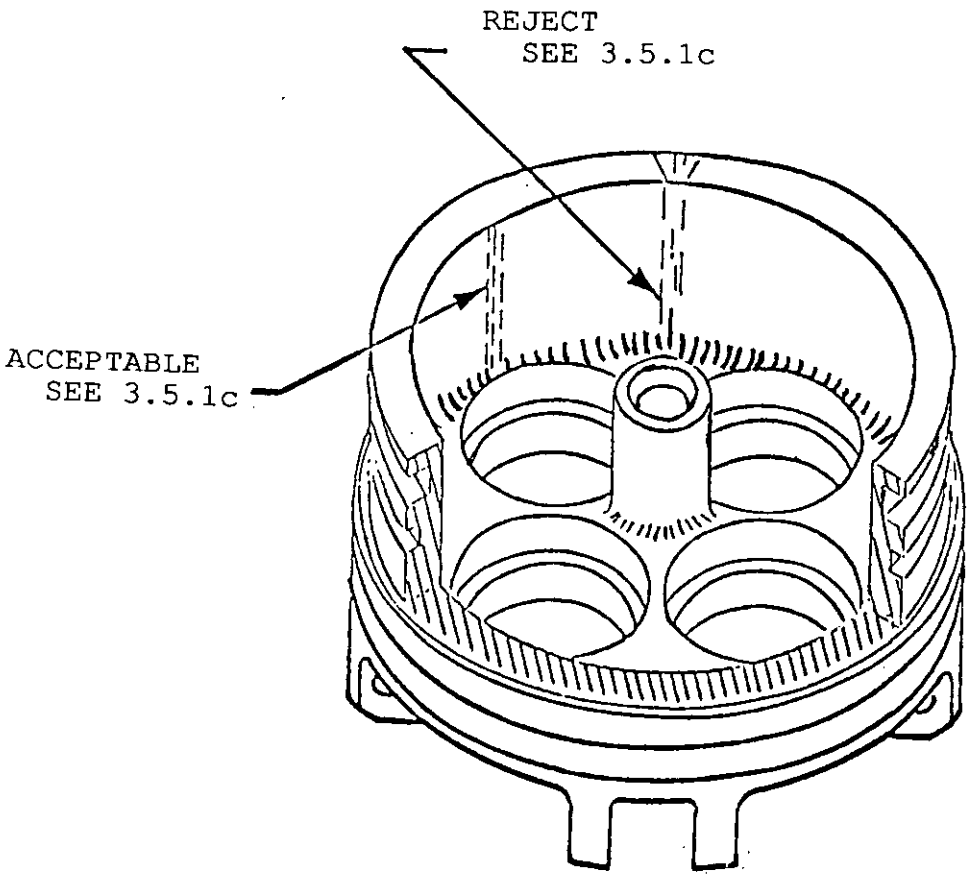


FIGURE 2. Discontinuities in piston skirt.

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3.8 Burst strength. The burst diaphragm (see drawing 9220795), when in the configuration of the nozzle plate and inert assembly (see drawing 1517510) shall rupture at a gage pressure of 250 lbf/in^2 (pounds force per square inch) $\pm$ 75 lbf/in^2. Pressurization by a fluid shall be at a uniform rate between a gage pressure of 30 lbf/in^2 and 60 lbf/in^2 from zero toward 325 lbf/in^2. There shall be no solid material separated from the assembly upon rupture.

3.9 Lead pull strength. The nozzle and fin assembly shall be capable of withstanding a 12 pound pull for 2 seconds between the nozzle plate and the forward connector of the ground lead and between the nozzle plate and the forward connector of the live lead, and between the nozzle plate, (with the cross head restrained from rearward movement), and the contact disc with the pin retainer disengaged from the fin blades.

3.10 Piston pressurization strength. The piston assembly shall be capable of withstanding a gage pressure of 2000 + 50, -0 lbf/in^2 applied to that portion of the assembly forward of the "O" ring groove in the piston. Each piston assembly shall be capable of withstanding the above pressure applied in the above manner for 2 seconds without pressure drop or without leakage either at the end of the piston between the piston body and the insulation or between the insulation and the conductor.

3.11 Ground lead continuity. The nozzle and fin assembly shall be not greater than a resistance of 0.10 ohms between the nozzle plate and the forward end of the ground lead.

3.12 Air tightness. The nozzle and fin assembly shall withstand without a leakage rate of 10-4 cubic centimeters per second (cc/sec) or greater, an air pressure of 10 lbf/in^2 applied to the inner part of the assembly. The emission of visible air bubbles using approved equipment for submersion in water shall be considered to indicate the 10-4 cc leakage rate.

3.13 Workmanship. All parts shall be fabricated and finished such as to ensure a high quality product which is uniform and in conformance with this specification. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, metal shavings, grease, rust, corrosion products and other foreign matter. All required markings shall be neat and sharply defined.

3.13.1 Metal parts. The metal parts used shall be free from defects, including all surface imperfections and discontinuities and all internal defects in the finished component resulting from flaws or lack of soundness in the materials used.

3.13.1.1 Surface imperfections. Surface imperfections include pits, seams, laps, tears, cracks, scale, blisters, clod shuts, and other defects of similar type.

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3.13.1.2 Internal defects. Internal defects include pipes, lamination, segregations, internal cracks, porosity, flakes, inclusions, and other defects of similar type.

3.13.2 Cleaning method. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

3.13.3 Surface coatings. Exterior surface coatings shall be continuous except for a few light scratches not exposing bare material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate if specified in the contract or order (see 6.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted. The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at a temperature of 16 to 32° Celsius (C), relative humidity from ambient to 95 percent and barometric pressure of facility ground.

4.3.1 Equipment lists. The contracting officer shall specify the equipment lists (EL) necessary for the pertinent drawings (see 6.2).

4.3.2 Lot formation. A lot shall consist of a homogeneous collection of units manufactured in one continuous operation by the same process, by the same manufacturer, using the same materials, and in accordance with the same drawings, specifications, or revisions.

4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.4.1 Submission. The contractor shall submit a first article sample to a Government approved facility as specified in the contract (see 6.3) for evaluation in accordance with the provisions of 4.5.1. The first article sample shall consists of 30 assemblies and 30 complete sets of parts, (that is, 30 each of every component and every assembly) which have been produced by the contractor or furnished by a supplier and which have been produced by the contractor production processes, procedures and equipment which will be used

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in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings, certificates of conformance for materials (see 4.1.2). A first article sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production or a period no greater than 90 days or whenever a change occurs in manufacturing, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.4.2 Rejection. If any assembly or component fails to comply with any of the requirements and inspections, the First Article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new first article sample or portion thereof. Until a first article sample is accepted the contractor is, in no way authorized by the Government to initiate regular production unless directed by the contracting officer.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in the applicable drawings, as specified herein, and as specified by the contracting officer (see 6.2).

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TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Material	3.2	4.6.1
Nozzle and fin assembly	3.3	4.8.13
Metal parts and other material	3.4	4.1.2, 4.4.1.1
Surface imperfections	3.4.1	4.1.2, 4.4.1.1
Internal defects	3.4.2	4.1.2, 4.4.1.1
Magnetic particle inspections	3.5	4.8.7
Cracks or cracklike indications	3.5.2	4.1.2
Pivot pin push test	3.6	4.8.11
Insulation integrity	3.7	4.8.14
Burst strength	3.8	4.8.8
Lead pull strength	3.9	4.10.13
Piston pressurization strength	3.10	4.8.6
Ground lead continuity	3.11	4.8.15
Air tightness	3.12	4.10.12
Workmanship	3.13	4.4.1.1, 4.6.1

4.5.1 Inspections to be performed. Assemblies and components shall be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the drawing.

4.6 Inspection provisions for production.

4.6.1 Lot formation. The term "inspection lot" as used throughout this specification is an essentially homogeneous collection of units or product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, drawing revision, specification and specification revision and complies with the provisions for submission of the product. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number. Unless otherwise approved by the contracting officer, the inspection

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lot size of end items deliverable under the contract shall be not less than 7,000 nor more than 50,000 units.

4.6.2 Examination. The examination specified herein shall be performed on inspection lots (see 4.6.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified shall be 100 percent). Sampling plans shall be in accordance with tables II through VII of this specification.

TABLE II. Sampling plan I.

Lot size	Sample size
2 to 50	All
51 to 500	50
501 to 1200	75
1201 to 10,000	116
10,001 to 35,000	135
35,001 to 150,000	170
150,001 to 500,000	200
500,001 and over	244
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE III. Sampling plan II.

Lot size	Sample size
2 to 32	All
33 to 280	32
281 to 500	48
501 to 3200	73
3201 to 10,000	86
10,001 to 35,000	108
35,001 to 150,000	123
150,001 to 500,000	156
500,001 and over	189
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE IV. Sampling plan III.

Lot size	Sample size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE V. Sampling plan IV.

Lot size	Sample size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE VI. Sampling plan V.

Lot size	Sample size
2 to 8	All
9 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38
10,001 to 35,000	46
35,001 to 150,000	56
150,001 and over	64
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE VII. Sampling plan VI.

Lot size	Sample size
2 to 5	All
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	40
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.7 Classification of defects. Each sample shall be tested for defects as specified in 4.7.1 through 4.7.19.

4.7.1 Contact disc (see drawing 1253129 covering a detail of drawing 1517508 and table VIII.

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TABLE VIII. Classification of defects, contact disc.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Plating missing or inadequate	Visual	III

4.7.2 Fin blade (see drawing 656926 covering a detail of drawing 1517508 and table IX).

TABLE IX. Classification of defects, fin blade.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Blade straightness (note 6a)	Gage	VI
102.	Twist (note 6b)	Gage	VI
103.	Blade symmetry (note 6c)	Visual	VI
104.	Protective finish missing or improper	Visual	VI
105.	Thickness of blade at hole, boss area	Gage	VI
106.	Crosshead bearing surface not coincident with center line of hole	Gage	VI
107.	Wall thickness around hole, minimum	Gage	VI
108.	Distance from center of hole to leading edge	Gage	VI

4.7.3 Spring pin (see MS 171590 covering detail of drawing 1517508 and table X).

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TABLE X. Classification of defects, spring pin.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Outside diameter	Gage	III

4.7.4 Cross head (see drawing 456909 covering a detail of drawing 1517508 and table XI).

TABLE XI. Classification of defects, cross head.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Both faces not perpendicular to piston hole	Gage	V
102.	Concentricity of piston hole to outside diameter	Gage	V
103.	Protective finish missing or improper	Visual	V

4.7.5 "O" ring (see drawing 650950 covering a detail of drawing 1517508 and table XII).

TABLE XII. Classification of defects, "O" ring of drawing 650950.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Nicks, cuts, surface defects or surface irregularities	Visual	III

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4.7.6 "O" ring (see drawing 650953 covering a detail of drawing 1517508 and table XIII).

TABLE XIII. Classification of defects, "O" ring for drawing 650953.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Diameter of cross section	Gage	III
102.	Nicks, cuts, surface defects or surface irregularities	Visual	III

4.7.7 Piston (glass bead, one end), (see drawing 9220796 covering a detail of 1517508 and table XIV). This paragraph shall be applied when the piston assembly (glass bead, one end) drawing 9220782 is supplied in the nozzle and fin assembly.

TABLE XIV. Classification of defects, piston (glass bead, one end).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Diameter of knurled section after knurling	Gage	VI
102.	Plating missing or improper	Visual	VI
103.	Head diameter	Gage	VI
104.	Body diameter	Gage	VI
105.	Cross head shaft diameter	Gage	VI
106.	Thread improper	Gage	VI
107.	Concentricity of head diameter with body diameter	Gage	VI

4.7.8 Piston assembly (glass bead, one end), (see drawing 9220782 covering a detail 1517508 and table XV). This paragraph shall be applied when this assembly is supplied in the nozzle and fin assembly.

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TABLE XV. Classification of defects, piston assembly (glass bead, one end).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical Major 101.	None defined Glass bead cracked, loose or missing	Visual/ Manual	III

4.7.9 Piston (glass bead, two ends), (see drawing 9220783 covering a detail of 1517508 and table XVI). This paragraph shall be applied when the piston assembly (glass bead, two ends) (drawing 9220787) is supplied in the nozzle and fin assembly.

TABLE XVI. Classification of defects, piston (glass bead, two ends).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical Major	None defined		
101.	Diameter of knurled section after knurling	Gage	V
102.	Head diameter	Gage	V
103.	Body diameter	Gage	V
104.	Cross head shaft diameter	Gage	V
105.	Thread improper	Gage	V
106.	Concentricity of head diameter with body diameter	Gage	V

4.7.10 Piston assembly (glass bead, two end), (see drawing 9220797 covering a detail of drawing 1517508 and table XVII). This paragraph shall be applied when this assembly is supplied in the nozzle and fin assembly.

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TABLE XVII. Classification of defects, piston assembly (glass bead, two ends).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Glass bead cracked, loose or missing	Visual/ Manual	III
102.	Plating missing or improper	Visual	III

4.7.11 Electrical lead live assembly. Prior to assembly of forward connector (see drawing 9220785 covering a detail of 1517508 and table XVIII).

TABLE XVIII. Classification of defects, electrical lead live assembly.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Overall length of assembly	Gage	III
102.	Part missing	Visual	III

4.7.12 Fin retainer and live lead assembly (see drawing 9234433 covering a detail of drawing 1517508 and table XIX).

TABLE XIX. Classification of defects, fin retainer and live lead assembly.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Overall length of assembly	Gage	III
102.	Contact disc not properly or securely fastened to fin retainer	Visual/ Manual	III

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4.7.13 Electrical lead ground before assembly of forward connector (see drawing 1253133 covering a detail of 1517508 and table XX).

TABLE XX. Classification of defects, electrical lead ground before assembly of forward connector.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Length of wire	Gage	III

4.7.14 Nozzle plate, before plating (see drawing 656925 covering a detail of 1517508 and table XXI).

TABLE XXI. Classification of defects, nozzle plate, before plating.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Diameter of nozzle part, maximum	Gage	III
102.	Concentricity of pitch circle to outside diameter of forward section of plate	Gage	III

4.7.15 Nozzle plate after plating - inserts assembled (see drawing 656925 covering a detail of drawing 1517508 and table XXII).

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TABLE XXII. Classification of defects, nozzle plate after plating.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Concentricity "O" ring groove diameter to outside diameter at forward section of plate	Gage	VII
102.	Concentricity of lockwire groove diameter to outside diameter at forward section of plate	Gage	VII
103.	Concentricity of bourrelet to outside diameter at forward section of plate	Gage	VII
104.	Concentricity of piston hole diameter at aft surface to outside diameter at forward section of plate	Gage	VII
105.	Perpendicularity of lockwire groove to axis of outside diameter at forward section of plate	Gage	VII
106.	Nicks, radial scratches, gouges, and spiral tool marks on forward face of plate and in "O" ring groove	Visual	VII
107.	Outside diameter of forward section of plate	Gage	VII
108.	"O" ring groove diameter	Gage	VII
109.	Location of fin pivot holes from center of bourrelet diameter	Gage	VII
110.	Diameter of fin pivot holes, minimum	Gage	VII
111.	Width of fin pivot slots, maximum	Gage	VII
112.	Thickness of material below lockwire hole	Gage	VII
113.	Outside diameter at fin pivot ears, minimum	Gage	VII
114.	Width of lockwire groove	Gage	VII

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TABLE XXII. Classification of defects, nozzle plate after plating (continued).

Categories	Defects	Method of inspection	Sampling plan (tables)
115.	Diameter of lockwire groove	Gage	VII
116.	Concentricity of piston hole diameter at aft surface with forward piston hole diameter	Gage	VII
117.	Forward piston hole diameter	Gage	VII
118.	Diameter of piston hole in aft face of plate	Gage	VII

4.7.16 Nozzle insert (see drawing 456916 covering a detail of drawing 1517508 and table XXIII).

TABLE XXIII. Classification of defects, nozzle inert.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Concentricity of nozzle throat with outside diameter of press fit section	Gage	V
102.	Concentricity of 0.500 radius with nozzle exit diameter	Gage	V
103.	Throat mismatch to 0.500 radius	Visual	V
104.	Outside diameter of press fit, minimum	Gage	V
105.	Concentricity of outside diameter of press fit with nozzle exit diameter	Gage	V

4.7.17 Nozzle plate and insert assembly before plating and assembly of burst diaphragm (see drawing 1517510 covering a detail of 1517508 and table XXIV).

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TABLE XXIV. Classification of defects, nozzle plate and insert assembly before plating and assembly of burst diagram.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Nozzle insert stop shoulder not fully and firmly seated against nozzle plate	Visual	VII

4.7.18 Diaphragm, burst (nozzle insert) (see drawing 9220795 covering a detail of drawing 1517508 and table XXV).

TABLE XXV. Classification of defects, diaphragm, burst (nozzle insert).

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Outside diameter, minimum	Gage	III
102.	Length of coin, minimum	Gage	III

4.7.19 Nozzle plate and insert assembly (see drawing 1517510 covering a detail of 1517508 and table XXVI). This paragraph shall be applied to Mark 4 nozzle and fin assemblies.

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TABLE XXVI. Classification of defects, nozzle plate and insert assembly Mark 4.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined		
Major			
101.	Nozzle throat diameter	Gage	III
102.	Plating missing or improper	Visual	III
103.	Burst diaphragm positioned improperly (to include off center and/or backward) or missing	Visual	III
104.	Perpendicularity of nozzle cone to lockwire groove "V" surface	Gage	III

4.7.19.1 Nozzle plate and insert assembly (see drawing 1517510 covering a detail of 1517508 and table XXVII). This paragraph shall be applied to Mark 40 nozzle and fin assemblies.

TABLE XXVII. Classification of defects, nozzle plate and insert assembly Mark 40.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical	None defined.		
Major			
101.	Nozzle throat diameter	Gage	V
102.	Plating missing or improper	Visual	V
103.	Burst diaphragm positioned improperly (to include off center and/or backwards or missing	Visual	V
104.	Height of forward end of scarf from forward face	Gage	V
105.	True position of plane of scarf	Gage	V
106.	Perpendicularity of nozzle cone to lockwire groove "V" surface	Gage	V

4.7.20 Nozzle and fin assembly (see drawing 1517508 and table XXVIII).

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TABLE XXVIII. Classification of defects, nozzle and fin assembly.

Categories	Defects	Method of inspection	Sampling plan (tables)
Critical			
1.	Assembly does not pass through launch tube simulator	Gage	100%
Major			
101.	Part missing	Visual	V
102.	Fin retainer not securely assembled	Visual/ Manual	V
103.	Contact disc, connector and lead not properly insulated	Visual	V
104.	Fin spacing unsymmetrical	Visual	V
105.	Terminal wire bent excessively	Visual	V
106.	Distance of uninsulated terminal wire exposed on aft end	Gage	V

4.8 Testing.

4.8.1 Fin blade tensile test. Each furnace (heat treat) lot of fin blades shall be sampled and tested for fin blade strength. Furnace (heat treat) lots shall be sampled and tested according to the following:

a. According to 4.8.1.1, until five consecutive lots have passed fin blade material verification. Thereafter, one lot in five furnace (heat treat) lots shall be randomly selected, and so tested.

b. All furnace (heat treat) lots which are not sampled and tested according to 4.8.1.1 shall be sampled and tested according to 4.8.1.2.

c. Rejection of any furnace (heat treat) lot shall require return to testing as in "a" above.

4.8.1.1 Fin blade material verification test [see drawing 656926 note 1.a (1) or (2)]. Three fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.10.1.1 using equipment in accordance with 4.9. Failure of one or more fin blades to meet the requirements of Note 1 of drawing 656926 shall cause furnace (heat treat) lot rejection. In the event that lot rejection criteria are reached due to no test (machine failure, specimen failing outside gage length, and so forth) each no test fin blade shall be replaced in the sample by another fin blade drawn from the heat treat lot.

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4.8.1.2 Fin blade pin pull test [see drawing 656926 note 1.a (3)]. Ten fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.10.1.2 using equipment in accordance with 4.9. Failure of one or more fin blades to meet the requirements of Note 1.a (3) on drawings 656926 shall cause furnace (heat treat) lot rejection. Failure of the fin blade to be pulled to be not less than the load attributable to the test equipment (such as pin failure) shall be considered a no test fin blade. In the event of more than one no test fin blade in the sample, additional fin blades shall be selected so that nine fin blades shall be validly tested.

4.8.2 Hardness test on spring pin (see drawing MS171590 covering a detail of 1517508). A sample consisting of 25 spring pins shall be taken from each lot of spring pins. The sample shall be tested as specified in 4.10.2 using equipment in accordance with 4.9. Failure of one or more specimens to indicate hardness within the specified limits shall cause rejection of the lot. This test is considered destructive and parts tested shall not be returned to the lot.

4.8.3 Vise test on spring pin (see MS171590 covering a detail of 1517508). A sample consisting of 125 spring pins shall be taken from each lot of spring pins. The sample shall be tested as specified in 4.10.3 using equipment in accordance with 4.9. Any pin in which a crack appears shall be considered defective. One or more defective spring pins in the sample shall cause rejection of the lot. This test is considered destructive and parts tested shall not be returned to the lot.

4.8.4 Compression test on cross head (see drawing 456909 covering a detail of 1517508). This paragraph shall apply when a powdered metal cross head is suppled in the nozzle and fin assembly. A sample consisting of 50 cross heads shall be taken from each lot of cross heads. The sample shall be tested as specified in 4.10.4 using equipment in accordance with 4.9. Failure of any cross head in the sample to meet the requirements of Note 8.a on drawing 456909 shall cause rejection of the lot.

4.8.5 Bending test on cross head (see drawing 456909 covering a detail of 1517508). This paragraph shall apply when a powdered metal cross head is supplied in the nozzle and fin assembly. A sample consisting of 50 cross heads shall be taken for each lot of cross heads. The sample shall be tested as specified in 4.10.5 using equipment in accordance with 4.9. Failure of any cross head in the sample to meet the requirements of Note 8.b on drawing 456909 shall cause rejection of the lot.

4.8.6 Pressure test on piston assembly (see drawing 9220782 or drawing 9220797 covering alternate details of drawing 9220785 for piston assembly glass bead, one end for piston assembly glass bead, two ends). This test shall be performed on the piston assembly dependant on the drawing to which the supplied assembly conforms. The sampling plan for this test shall be in accordance with table VI (sampling plan V). This test shall be performed as

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specified in 4.10.6 using equipment in accordance with 4.9. Any piston assembly which does not meet the requirements specified in 3.10 shall be considered defective.

4.8.7 Magnetic particle inspection of nozzle plate (drawing 656925 covering a detail of 1517510 (see 3.5). Each lot of nozzle plates shall be subjected to 100 percent magnetic particle inspection as specified in 4.10.7 using equipment in accordance with 4.9. Any nozzle plate failing to meet the requirements specified in 3.5 shall be considered defective, and the lot shall be rejected.

4.8.8 Burst test on nozzle plate and insert assembly (drawing 1517510 covering a detail of 1517508) (see 3.8). A sample of 20 nozzle plate and insert assemblies shall be taken from each nozzle plate and insert assembly lot. The sample shall be tested as specified in 4.10.8 using equipment in accordance with 4.9. A burst diaphragm which fails to rupture as specified in 3.8 shall be counted as one defect. (Therefore, each nozzle plate and insert assembly in the sample has a potential of four defects.) If there is one or more defect in the sample the lot shall be rejected.

4.8.9 Piston free movement test on nozzle and fin assembly with the fin retainer and contact disc disengaged from the fin blades (see drawing 1517508). The sampling plan for this test shall be in accordance with table IV (sampling plan III). This test shall be performed as specified in 4.10.9 using equipment in accordance with 4.9. Any nozzle and pin assembly with the fin retainer and contact disc disengaged from the fin blades in which the piston will not freely move 0.5 inch to the rear and return forward shall be considered defective.

NOTE: This test shall be performed in such a manner that the piston movement restriction of Note 2 BUORD drawing 1517508 is maintained.

4.8.10 Fin blade free fall test on nozzle and fin assembly with the fin retainer and contact disc disengaged from the fin blade (see drawing 1517508). The sampling plan for this test shall be in accordance with table IV (sampling plan III). This test shall be performed as specified in 4.10.10 using equipment in accordance with 4.9. Any nozzle and fin assembly with the fin retainer and contact disc disengaged from the fin blades on which any fin blade does not fall of its own weight when tested shall be considered defective.

4.8.11 Fin pivot pin push test on nozzle and fin assembly (see drawing 1517508). The sampling plan for this test shall be in accordance with table IV (sampling plan III). This test shall be performed as specified in 4.10.11 using equipment in accordance with 4.9. Any nozzle and fin assembly in which any spring pin moves when tested shall be considered defective.

4.8.12 Air test on nozzle and fin assembly (see drawing 1517508) (see 3.12). The sampling plan for this test shall be in accordance with table II

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(sampling plan I). This test shall be performed as specified in 4.10.12 using equipment in accordance with 4.9. Any nozzle and fin assembly which indicates a leakage rate greater than the requirement shall be considered defective. This test may be performed prior to complete assembly of all parts provided the nozzle plate and insert assembly (drawing 1517510), "O" ring (drawing 650950), "O" ring (drawing 650953), cross head (drawing 456909), nut self locking (drawing MS 21044-N3), and at least the piston assembly lead live assembly have been assembled to one another to form the units sampled for test.

NOTE: This test shall be performed in such a manner that the piston movement restriction of Note 2 BUORD drawing 1517508 is maintained.

4.8.13 Lead pull test on nozzle and fin assembly, with the fin retainer and contact disc disengaged from the fin blades (drawing 1517508) (see 3.9). The sampling plan for this test shall be in accordance with table III (sampling plan II). This test shall be performed as specified in 4.10.13 using equipment in accordance with 4.9. If the electrical lead live assembly or the ground lead exhibit separation or relative movement between any of the components, the assembly shall be considered defective.

4.8.14 Insulation integrity test on nozzle and fin assembly (drawing 1517508) (see 3.7). The sampling plan for this test shall be in accordance with table II (sampling plan I). The test shall be performed as specified in 4.10.14 using equipment in accordance with 4.9. Any nozzle and fin assembly which indicates less than the required resistance shall be considered defective.

4.8.15 Electrical continuity test on ground lead of nozzle and fin assembly (drawing 1517508) (see 3.11). The sample plan for this test shall be in accordance with table II (sampling plan I). This test shall be performed as specified in 4.10.15 using equipment in accordance with 4.9. Any nozzle and fin assembly, in which the resistance in the ground circuit is indicated as exceeding that specified shall be considered defective.

4.8.16 Electrical continuity test of live lead of nozzle and fin assembly (drawing 1517508). The sampling plan for this test shall be in accordance with table II (sampling plan I). This test shall be performed as specified in 4.10.16 using equipment in accordance with 4.9. Any nozzle and fin assembly, in which the resistance in the live lead is indicated as greater than specified in Note 14 on drawing 1517508 shall be considered defective.

4.8.17 Electrical continuity test on spring pin (drawing 1517508). The sampling plan for this test shall be in accordance with table II (sampling plan I). This test shall be performed as specified in 4.10.17 using equipment in accordance with 4.9. Any nozzle and fin assembly in which the resistance between the inside diameter of the spring pin and the bourrolet is indicated as greater than specified in Note 20 on drawing 1516508 shall be considered defective.

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4.9 Inspection equipment. Equipment tabulation numbers ET 1517508-1 and -2 identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with instructions provided by the contracting officer.

4.9.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.10 Test methods and procedures.

4.10.1 Fin blade tensile test. (see 4.8.1).

4.10.1.1 Fin blade material verification test (see 4.8.1.1). A specimen shall be prepared from each fin blade in the sample. Each specimen shall be pulled in accordance with tension test method 211.1 of FED-STD-151.

4.10.1.2 Fin blade pin pull test (see 4.8.1.2). Each fin blade in the sample shall be subjected to a pull force to the leading edge. The aft portion of the fin blade shall be gripped using suitable jaws which shall not grip any portion of the fin blade 2 inches or more from the aft end of the fin blade. The application of force shall be by a pin through the hole in the fin blade having a diameter of 0.187 $\pm$ .002 inches. The pin shall be supported on both sides of the fin blade with a distance between supports of 0.3715, + 0.0010, - 0 inches. The force shall be applied until failure of the fin blade occurs or until a load force of 3,000 pounds is reached. Observation shall be made of the load at failure or 3,000 pounds whichever condition is met. Observation shall be made of the load force at failure.

NOTE: Pin hardness of approximately Rockwell 57 is recommended.

4.10.2 Hardness test on spring pin (see 4.8.2). A specimen shall be prepared from each spring pin in the sample. Each specimen shall be prepared and tested in accordance with paragraph 5.2 of AMS 7207 and FED-STD-151, Method 243.1.

4.10.3 Vise test on spring pin (see 4.8.3). Each spring pin in the sample shall be placed in a vise and the gap in the spring pin squeezed closed. Each spring pin shall be removed and examined for cracks.

4.10.4 Compression test on cross head (see 4.8.4). Each cross head in the sample shall be subjected to a compression load of 1,000 pounds, applied alternately at the outer ends of each set of diametrically opposite arms. Deformation shall be determined by plug gaging the 0.1076 + 0.0010 - 0.000

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dimension after the load is removed. There shall be no visible cracks or breaks.

4.10.5 Bending test on cross head (see 4.8.5). Each cross head in the sample shall be subjected to a bending load of 500 pounds applied at the center of the cross head by not larger than a 0.217 inch diameter ram, with the cross head supported alternately by supports under each set of diametrically opposite arms. The supports shall be not less than 0.950 inches apart. Visual observation shall be made for permanent deformation of the cross head.

4.10.6 Pressure test on piston assembly (see 4.8.6). Each piston assembly in the sample shall be held in such a way as to subject that portion of the assembly forward of the "O" ring groove to a gage pressure of 2000 + 50, -0 lbf/in^2 for at least 2 seconds. Observation shall be made for leakage or pressure drop.

4.10.7 Magnetic particle inspection of nozzle plate (see 4.8.7). Each nozzle plate shall be magnetic particle inspected in accordance with MIL-STD-1949. For surface indications, the penetrant method of inspection in accordance with MIL-STD-6866 may be used.

4.10.8 Burst test on nozzle plate and insert assembly (see 4.8.8). The burst test shall be performed on each burst diaphragm in each nozzle plate and insert assembly in the sample. Each burst diaphragm shall be individually subjected to a pressure applied to the forward slide of the burst diaphragm at a rate in accordance with 3.8. Observation shall be made for the pressure at rupture and each burst diaphragm shall be examined for the presence of all material.

4.10.9 Piston free movement test (see 4.8.9). Each nozzle and fin assembly in the sample with the fin retainer and contact disc disengaged from the fin blades shall be partially functioned by moving the piston 0.5 inches aft. The piston shall then be returned to the forward position. Do not move the piston more than 0.625 inches. Excessive movement will damage serrations on the piston.

4.10.10 Fin blade free fall test (see 4.8.10). Each nozzle and fin assembly in the sample, with the fin retainer and contact disc disengaged from the fin blades shall be supported on the "R" surface of the nozzle plate. The fin blades shall be raised to a position which is not less than 60° above the horizontal. The fins shall be released and allowed to fall of their own weight.

4.10.11 Fin pivot pin push test (see 4.8.11). Each nozzle and fin assembly in the sample shall be held with the nozzle plate positively positioned. A static load of 20 pounds shall be applied alternately to each end of each spring pin. Observation shall be made for movement during application of the load.

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4.10.12 Air test (see 4.8.12). Each nozzle and fin assembly shall be subjected to 10 lbf/in^2 air pressure internally for at least 10 seconds. Observation shall be made for bubbles if a wet submergence test is utilized; however, parts must be thoroughly dried after completion of the test. If a dry method of test is used, observation shall be made for the leakage rate. The cross head must be restrained in the forward position during testing.

4.10.13 Lead pull test (see 4.8.12). Each nozzle and fin assembly in the sample with the fin retainer and contact disc disengaged from the fin blades shall undergo the following: The nozzle plate shall be held with the cross head secured from movement. Alternately, the forward connector on the live lead, the contact disc on the live lead, and the forward connector on the ground lead shall have a 12 pound pull applied for not less than 2 seconds. Observation shall be made for relative movement between any part of any lead and the nozzle plate.

4.10.14 Insulation integrity test (see 4.8.14). Each nozzle and fin assembly in the sample shall have a 100 volt potential applied to measure the resistance between the contact disc and the bourrelet of the nozzle plate.

4.10.15 Electrical continuity test on ground lead (see 4.8.15). The resistance of each nozzle and fin assembly in the sample shall be measured between the bourrelet on the nozzle plate and the forward connector on the ground lead.

4.10.16 Electrical continuity test on live lead (see 4.8.16). The resistance of each nozzle and fin assembly in the sample shall be measured between the contact disc and the forward connector on the electrical lead live assembly.

4.10.17 Electrical continuity test on spring pin (see 4.8.17). The resistance of each nozzle and fin assembly in the sample shall be measured between the inside diameter of the spring pin and the bourrelet of the nozzle plate.

4.11 Ammunition data. As specified in the contract or order (see 6.2) data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167. Lots shall be numbered in accordance with MIL-STD-1168.

4.12 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of Section 5.

5. PACKAGING

5.1 Packing. Packing shall be level C. Forty nozzles and fin assemblies shall be packed in a PPP-B-636, type C, style CSSC, class Domestic/Fire Retardant, variety SW, grade 275 fiberboard box. Each box shall

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be fitted with half-slotted partitions assembled to form 40 individual cells in an 8 by 5 arrangement. Partitions shall be constructed of the same fiberboard as the box except "B" flute, grade 200 board shall be used. Each cell shall hold one assembly in an upright position. Provision shall be made to protect the forward end of the live lead wire which protrudes out of the piston. Boxes shall be closed by application of 2 inch wide tape, PPP-T-45, type III grade B, tape over the full length of the top and bottom seams of the box with approximately 2 inch minimum extension of tape onto the end panels and around box corners.

5.2 Palletization. Boxes shall be palletized for shipment. Pallets shall conform to NN-P-71, type II or III, size 1 or 2. The gross weight for each pallet shall be not greater than 2,000 lbs. Not less than 4 straps, 2 straps around the width and length of the pallet load, shall secure the load. Strapping shall conform to QQ-S-781, type I or IV, 0.625 x .020 inch. Provision shall be made to prevent strapping from cutting into the fiber boxes without tension by use of nominal 1 x 4 wooden boards placed under the straps or by use of edge protectors.

5.3 Marking. Each box and pallet load shall be marked in accordance with MIL-STD-1169.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nozzle and fin assembly in this specification is intended to be used for a 2.75 inch rocket motor to propel a Mark 4 or a Mark 40 rocket.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Which type is required (see 1.2)

d. When first article is required (see 3.1 and 4.4)

e. Certificate of findings and inspection of product (see 4.1.2)

f. Which equipment lists are required (see 4.3.1)

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g. Other sample quantities required (see 4.8.1.1, 4.8.1.2, 4.8.2 and 4.8.3)

h. When ammunition data cards are required (see 4.11).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Cross-reference. The item listed below is changed as specified:

9241782	Spring Pin	to	MS171590

6.5 Government loaned property. When equipment lists specify government owned test fixture, the contracting officer will arrange for the loan of this equipment.

6.6 Re-use of wooden pallets. It is recommended that re-use of the wooden pallets used for packaging and packing be authorized.

6.7 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.8 Subject term (keyword) listing.

Engine
Missile non-guidance control element
Stabilizing device

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review Activity:
Air Force - 70

Preparing Activity:
Army - MI

(Project No. 1340-A832)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-60445C	910603

3. DOCUMENT TITLE
FIN AND NOZZLE ASSEMBLY FOR ROCKET MOTOR 2.75 INCH MARK 4 AND MARK 40 PARTS FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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